

Business Requirements Document — FinTech Platform

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1. Executive Summary

FinTech Platform is a reference digital-banking codebase used as a hands-on simulation workspace: a candidate joins as if onto a real platform team building core banking infrastructure. The repository scaffolds a full bank (customer, accounts, cards, lending, trading, fraud, ML, data, security) but currently ships one fully working vertical slice — onboarding a customer, opening an account, and transferring money between accounts, with every balance backed by a real double-entry general ledger. The business need is a realistic, safely-scoped environment where engineers can be evaluated on and trained in the skills real banking-platform work demands: service boundaries, financial correctness, and safe extension of an existing system, without the cost or risk of a production banking environment.

2. Business Objectives

1. Provide a technically credible banking-domain workspace that reflects how a real digital bank is actually built (microservices, database-per-service, an auditable ledger), not a toy CRUD app.
2. Give candidates/engineers a structured, day-by-day body of work (tickets) that mirrors real platform priorities — fix defects, hit security and compliance points (KYC, PCI-DSS), harden reliability, and extend the product (cards, lending, events) — over a 30-day simulation.
3. Pair the hands-on ticket work with a DSA practice track and a knowledge-check quiz pool so technical readiness is assessed alongside real system work.

3. Scope

3.1 In Scope

- The five implemented services (customer, account, general-ledger, internal-transfer, api-gateway) and the web-banking demo UI, as the base the 30-day ticket program builds on.
- 30 daily tickets spanning bug fixes, feature work, security/compliance hardening, performance, reliability, incident response, tech-debt, and deployment — extending the base slice into cards, lending, event messaging, and authentication.
- A 30-question DSA practice set aligned to problem types commonly asked in IBM technical interviews.
- A quiz question pool covering both general software-engineering fundamentals commonly asked at IBM and domain-specific concepts from this codebase (double-entry bookkeeping, idempotency, database-per-service, circuit breakers).

3.2 Out of Scope

- Building out the untouched scaffold areas not reached by the 30-day ticket set (trading, fraud, ML platform, full Kubernetes/Terraform infrastructure, multi-currency FX).
- Any connection to real banking rails, real customer data, or real payment networks — this is a simulated environment only.
- Production-grade authentication/authorization beyond the single hardening ticket included in the 30-day set (day 23).

4. Stakeholders

Role	Name/Team	Interest
Sponsor	Platform Engineering Leadership	Wants a repeatable, realistic way to assess and train engineers on banking-platform skills
Product Owner	OFFICE iQ Content Team	Owns the workspace catalog entry and its content pack (tickets, DSA, quiz)
Candidates / Engineers	Simulation participants	Complete the 30-day ticket program, DSA problems, and quizzes

5. Business Requirements

ID	Requirement	Priority
BR-1	The workspace must present a working, runnable banking system on day 1 (<code>docker compose up --build</code>), not just documentation	Must
BR-2	Every simulation day (1-30) must have exactly one ticket, each traceable to a real file/service in the codebase	Must
BR-3	The ledger's double-entry invariant (debits equal credits) must never be weakened by any ticket in the program	Must
BR-4	The program must include at least one ticket each for security, incident response, and compliance-adjacent work (KYC, PCI-DSS)	Should
BR-5	DSA practice content must reflect problem types genuinely representative of IBM technical interviews	Should
BR-6	Quiz content must cover both general CS/Java fundamentals asked in IBM interviews and this codebase's specific architecture	Should

6. Assumptions & Constraints

- **Assumptions:** Participants have Docker installed and can run a multi-service local stack; participants have baseline Java/Spring Boot and React/TypeScript familiarity going in.
- **Constraints:** The backend Java services were authored without Maven Central access and were not compiled before handoff (see README) — day 2's ticket exists specifically to address this. Content-pack hard limits apply: at most 30 tickets (one per day), at most 30 DSA problems, at most 200 quiz questions.

7. Success Metrics / KPIs

- 100% of the 30 simulation days have a ticket that references a real, existing part of the codebase.
- All 30 DSA problems and all quiz questions pass content-pack validation on upload with zero 422 errors.
- Candidates completing the 30-day program can articulate the ledger's double-entry invariant and at least one reliability pattern (idempotency, retries/circuit breaker, outbox) introduced during the program.

8. Risks

Risk	Impact	Mitigation
Backend services fail to compile locally (no Maven Central access at authoring time)	High	Day 2 ticket dedicates time to fixing build issues before other work begins
Candidates skip foundational days and jump to extension tickets (cards/lending), missing core ledger concepts	Medium	Tickets are ordered by day and each extension ticket depends on concepts introduced earlier in the sequence
DSA/quiz content drifts from what IBM interviews actually ask over time	Low	Content is grounded in current published interview-question compilations and should be refreshed periodically

9. Glossary

- **Vertical slice** — the one fully-built, end-to-end path (onboarding → account → transfer) through an otherwise scaffolded system.
- **Double-entry bookkeeping** — an accounting method where every transaction records equal, offsetting debit and

credit entries.

- **KYC** — Know Your Customer; identity verification required before a financial account can be opened.
- **Idempotency key** — a client-supplied token letting a server safely recognize and dedupe a retried request.

10. Approval

Name	Role	Date
Platform Engineering Leadership	Sponsor	2026-08-24
OFFICE iQ Content Team	Product Owner	2026-08-24